The Mixed-Boundary Poisson Problem: One-Sided vs Ghost-Point Closure, Measured and Compared

Team O: Zihan Xu Beixue Liang Yumo Li
Group project — Numerical Methods for PDEs (Project 2, Direction 2)

November 7, 2026

Abstract

Steady heat conduction in a device whose walls impose different kinds of thermal contact — a prescribed temperature, an imposed flux, a convective face — leads to an elliptic problem whose *boundary closure is part of the numerical method*. We solve the Direction-2 model, $-\Delta u = f$ on the unit square with Dirichlet data on three sides (four corners included) and, on the open right edge, either a Neumann condition $\partial_n u = q$ or a Robin condition $\partial_n u + \beta u = r$ ($\beta = 2$), with a from-scratch five-point sparse finite-difference assembly and the two prescribed closures: a first-order one-sided derivative row that *replaces* the PDE row, and a second-order ghost-point elimination that *keeps* it.

Three results carry the report. First, both closures do exactly what Taylor expansion says they should: against the manufactured solution $u^* = e^{x+y}$ the ghost variants fit orders 2.04/2.00 (Neumann) and 2.05/2.00 (Robin), the one-sided variants fit 1.06/1.02 and 1.05/1.01, and hand substitution of u^* into every non-standard row reproduces the predicted defect constants to three digits. Second, the interesting subtlety is real and measurable: the ghost row's *local* defect is $O(h)$, the same order as the one-sided row's, yet it delivers global order two because the defect lives in a boundary layer of width h, while the one-sided defect *is* the boundary error. Third, the closure choice is a *feasibility* choice, not a cosmetic one: boundary error $L^\infty \le 10^{-4}$ needs $N \approx 47$ with the ghost closure but $N \approx 8.6 \times 10^3$ (about 7×10^7 unknowns) with the one-sided row. Because both mixed assemblies are nonsymmetric, conjugate gradients is off the table; sparse LU and BiCGSTAB+ILU (39 iterations at relative residual 10^{-10}) take its place, and a tolerance sweep shows the reported orders are discretisation rates, not solver artefacts.

Every number quoted below is produced by `code/run_all.py` in one command, and every table body is generated from that run, so the report cannot drift from the code. The main limitation is stated up front: the optional pure-Neumann depth variant converges at L^∞ order $1.84 < 2$; we can attribute the gap to the corner rows but cannot yet derive the rate.

AI assistance. This report was produced with substantial AI assistance. A full log is given in Appendix A. Per the project brief, all three members can explain and defend the mathematics, the implementation and the conclusions in the oral viva without AI assistance.

Keywords: elliptic boundary value problem; five-point stencil; ghost-point elimination; method of manufactured solutions; sparse solvers

Contents

1 Introduction and problem statement **3**
1.1 The task . . . 3
1.2 The mixed-boundary Poisson problem . . . 3
1.3 Objectives . . . 3

2 Discretisation: the scheme and the two closures **4**
2.1 The interior scheme . . . 4
2.2 Closure 1 — the one-sided derivative row (replaces the PDE row) . . . 4
2.3 Closure 2 — ghost-point elimination (keeps the PDE row) . . . 4

2.4 What limits the global order — and how we measured it 5
2.5 Matrix structure 5

3 Implementation **5**
3.1 Algorithm 5
3.2 Code structure 6
3.3 Design decisions 6

4 Verification protocol **7**

5 Numerical experiments **10**
5.1 Error geometry: what the order table cannot show 10
5.2 Convergence orders 11
5.3 Matrix structure and the solver race 12
5.4 Cost vs accuracy: the closure choice is a feasibility choice 13
5.5 Falsification: can the study detect a bug? 14
5.6 Depth item: the pure-Neumann problem 14

6 Discussion **15**

7 Conclusions **16**

Appendix A — AI Transparency Log **17**

A AI Transparency Log **17**

Appendix B — Individual Contribution Statements **19**

B Individual Contribution Statements **19**

Appendix C — Problem Definition Sheet and Pre-computation Record **21**

C Problem Definition Sheet and Pre-computation Record **21**

1 Introduction and problem statement

1.1 The task

The project brief asks for a finite-difference solution of a Poisson problem with *mixed* boundary data: Dirichlet conditions on three sides of the unit square, and on the remaining side a choice of Neumann or Robin data closed in one of two prescribed ways. The brief is specific about what makes this interesting: the two right-edge closures differ by one line of the method, so any difference in accuracy, matrix structure or cost is attributable to that line alone — and the report must quantify it.

1.2 The mixed-boundary Poisson problem

Let $\Omega = (0,1)^2$. With the sign convention that the discrete operator is $-\Delta$ (positive diagonal),

$$-\Delta u = f \quad \text{in } \Omega, \tag{1}$$

with the boundary split $\partial\Omega = \Gamma_D \cup \Gamma_B$, $\Gamma_D = \{x = 0\} \cup \{y = 0\} \cup \{y = 1\}$ — all four corners belong to Γ_D, which removes any ambiguity about which discrete row assembles there — and the open right edge $\Gamma_B = \{1\} \times (0,1)$ carrying one of

$$\text{(Neumann)} \quad \partial_n u = q, \qquad\qquad \text{(Robin)} \quad \partial_n u + \beta u = r, \quad \beta = 2, \tag{2}$$

where n is the outward normal, so on $x = 1$, $\partial_n u = u_x(1, y)$. The physical reading is direct: a Dirichlet wall is a wall at a known temperature, a Neumann face is an imposed flux, and a Robin face is Newton's law of cooling, $-\alpha\, \partial_n u = h_c(u - u_\infty)$, which after division by the heat-transfer coefficient is exactly $\partial_n u + \beta u = r$ [1].

Following the problem pack we use the manufactured solution

$$u^*(x, y) = e^{x+y}, \qquad f = -\Delta u^* = -2e^{x+y}, \qquad g = u^*|_{\Gamma_D}, \tag{3}$$

and on Γ_B,

$$q(y) = u^*_x(1, y) = e^{1+y} > 0, \qquad r(y) = q(y) + \beta u^*(1, y) = 3e^{1+y}. \tag{4}$$

Two structural facts make this a clean test bed, and we use both throughout.

(i) The mixed problem is uniquely solvable. The only harmonic function with zero Dirichlet data on three sides and zero normal derivative on the fourth is $u \equiv 0$, so the pure-Neumann solvability constraint does not appear in the mixed case. (The depth experiment of Section 5.6 deliberately removes the Dirichlet sides and has to deal with it.)

(ii) The two closures differ by design, not by tuning. The brief prescribes both: a first-order one-sided derivative row that replaces the PDE row, and a second-order ghost-point elimination that keeps it. Nothing else in the method changes between the variants, so the convergence tables measure the closure and nothing but the closure.

1.3 Objectives

Concretely, we set out to answer five questions.

Q1. Do the two closures achieve the orders Taylor expansion predicts (1 for one-sided, 2 for ghost) — and can we measure the row defects themselves, not just the global rates?

Q2. Why does the ghost row converge globally at order two when its *local* defect is only order one?

Q3. What does the mixed boundary do to the matrix — and which solvers are actually justified?

Q4. At a matched accuracy target, what does each closure cost, and is the difference cosmetic or a matter of feasibility?

Q5. Can the verification machinery detect a plausible-looking wrong implementation — and what happens when the Dirichlet sides are removed entirely (the pure-Neumann depth item)?

2 Discretisation: the scheme and the two closures

2.1 The interior scheme

The nodal grid is $x_i = ih$, $y_j = jh$, $h = 1/N$, $i, j = 0, \dots, N$. Unknowns are the interior nodes $1 \le i, j \le N-1$ together with the right-edge nodes $i = N$, $1 \le j \le N-1$; every Dirichlet node carries index -1 and is eliminated, its value moved to the right-hand side with its actual coefficient. The ordering is row-major in x (documented in the `fdm_mixed.py` header). For a node with four active neighbours the five-point stencil for $-\Delta$ gives the row

$$\frac{4U_{i,j} - U_{i-1,j} - U_{i+1,j} - U_{i,j-1} - U_{i,j+1}}{h^2} = f_{i,j}, \tag{5}$$

and any neighbour on Γ_D contributes g/h^2 to the right-hand side. The matrix is assembled directly into sparse COO/CSR form (`scipy.sparse`); it is never built dense and converted.

Taylor expansion of (5) at (x_i, y_j) gives the leading truncation term

$$\tau_{\text{int}} = -\frac{h^2}{12}\left(u^*_{xxxx} + u^*_{yyyy}\right) + O(h^4), \tag{6}$$

so the interior scheme is second-order accurate — which $u^* = e^{x+y}$ satisfies on $\bar{\Omega}$. We do not just quote this: substituting sampled u^* into (5) at $(0.5, 0.5)$, $h = 1/64$, measures the defect -0.4531 against the predicted $-e^1/6 = -0.4530$ (Table 6).

2.2 Closure 1 — the one-sided derivative row (replaces the PDE row)

At a right-edge unknown $U_{N,j}$ the brief's first closure replaces the PDE row by the first-order derivative row

$$\text{(Neumann)} \qquad \frac{U_{N,j} - U_{N-1,j}}{h} = q_j, \tag{7}$$

$$\text{(Robin)} \qquad \frac{U_{N,j} - U_{N-1,j}}{h} + \beta U_{N,j} = r_j. \tag{8}$$

Taylor expansion at $x = 1$ gives the row defect $\tau_1 = -\frac{h}{2}u^*_{xx}(1, y) + O(h^2) = -\frac{h}{2}e^{1+y} + O(h^2)$: at $y = 0.5$, $h = 1/64$, measured -2.229 against the predicted -2.241. Two structural facts follow. The row scales like h^{-1}, unlike the h^{-2} PDE rows — the algebraic trace of replacing a PDE row by a derivative row. And because the row *replaces* the PDE row, its $O(h)$ defect measures the solution error at the boundary nodes directly, and that error then pollutes the domain: the global order is one.

2.3 Closure 2 — ghost-point elimination (keeps the PDE row)

The second closure introduces the conceptual value $U_{N+1,j}$ outside the domain and imposes the *centred* condition

$$\frac{U_{N+1,j} - U_{N-1,j}}{2h} = q_j \quad \text{or} \quad \frac{U_{N+1,j} - U_{N-1,j}}{2h} + \beta U_{N,j} = r_j, \tag{9}$$

hence $U_{N+1,j} = U_{N-1,j} + 2h(q_j - \beta U_{N,j})$ (with $\beta = 0$ in the Neumann case). Substituting into the five-point equation at $(1, y_j)$ eliminates the ghost value completely:

$$\frac{(4 + 2\beta h)U_{N,j} - 2U_{N-1,j} - U_{N,j-1} - U_{N,j+1}}{h^2} = f_{N,j} + \frac{2\,\text{data}_j}{h}, \tag{10}$$

with $\text{data}_j = q_j$ (Neumann) or r_j (Robin). Two things are visible in the row and both are measured: the inward coupling $U_{N-1,j}$ is *doubled*, and the centred boundary condition brings its own $O(h^2)$ defect ($\frac{h^2}{6}u^*_{xxx}$, measured 0.747 vs. 0.747 predicted) into the row at size $2h \cdot O(h^2)/h^2 = O(h)$: $\tau_2 = \frac{h}{3}u^*_{xxx}(1, y) + O(h^2)$, measured 1.482 against the predicted 1.494 at $h = 1/64$, $y = 0.5$ (Table 6).

2.4 What limits the global order — and how we measured it

Here is the subtlety that answers Q2, and it is the most instructive finding of the project. Closure 2's local row defect is $O(h)$ — the same order as Closure 1's — yet its global convergence order is two while Closure 1 delivers one. The resolution is geometric. The ghost row's $O(h)$ defect lives only on the $O(1)$ right-edge rows, i.e. in a boundary layer of width h, and the two-point normal difference constrains $e_N - e_{N-1} = O(h^3)$; the global effect is $O(h^2)$ [1]. Closure 1 replaces the PDE row entirely, so its defect *is* the boundary error, and the global order is one.

We treat this as a claim to test rather than a licence to assume a slope: the measured orders are in Tables 2–5, the row defects are in Table 6, and the falsification suite of Section 5.5 shows the refinement study can detect a wrong implementation.

2.5 Matrix structure

Both mixed assemblies are **nonsymmetric**: the derivative rows have a different scaling from interior PDE rows, and the ghost elimination doubles the inward coupling. Measured, $\left\|A - A^T\right\| / \|A\|$ falls from 1.6×10^{-2} at $N = 16$ to 2.0×10^{-3} at $N = 128$ (the boundary rows become a vanishing fraction of all rows, but never zero), and the one-sided assembly is more asymmetric still (1.5×10^{-1} at $N = 16$). This is the answer to Q3: the continuous operator is self-adjoint, but the mixed boundary rows destroy the symmetry, so conjugate gradients is **not** justified and we use sparse LU or nonsymmetric Krylov methods (BiCGSTAB + ILU). Dirichlet elimination leaves non-zero row sums wherever an entry was moved to the right-hand side, while the pure-Neumann assembly of Section 5.6 has *exactly* zero row sums (Table 9). The conditioning estimate grows like $h^{-2.00}$ (ghost) and $h^{-1.97}$ (one-sided), the one-sided matrix being $\approx 4\times$ worse at the same N.

3 Implementation

3.1 Algorithm

The core routine maps grid points to vector indices, inserts the stencil row for every interior node, applies the boundary closure on Γ_B, and solves with a sparse direct or an iterative solver to a stated tolerance.

Algorithm 1 Assembly of A_h and the right-hand side (mixed-BC Poisson)

Input: mesh parameter $h = 1/N$, closure $\in$ {onesided, ghost}, BC $\in$ {Neumann, Robin}, data f, g, q, r
Output: approximate solution u_h

```
Map interior nodes (i, j), 1 ≤ i, j ≤ N − 1, and right-edge nodes (N, j) to vector indices k; Dirichlet nodes get k = −1
for each interior grid point do
    Insert the five-point stencil row (5) into A_h (COO/CSR); for each Dirichlet neighbour add g/h^2 to b_k
end for
for each right-edge grid point (N, j) do
    if closure = onesided then
        Replace the row by the derivative row (7) or (8)
    else
        Substitute the ghost relation into the PDE row: doubled west coupling −2/h^2, RHS +2 data_j/h (10)
    end if
end for
return spsolve(A_h, b_h), or BiCGSTAB+ILU to relative residual tol ∼ Ch^2 (Section 4)
```

3.2 Code structure

The repository separates data, assembly, solvers, experiments and tests so that each can be checked independently.

File	Purpose
`code/manufactured.py`	Manufactured data u^*, f, g, q, r; forcing self-checks
`code/fdm_mixed.py`	Sparse assembly of A_h and b_h: index map, Dirichlet elimination, both closures
`code/solvers_iterative.py`	Jacobi / Gauss–Seidel / SOR / BiCGSTAB+ILU with residual stopping
`experiments/exp01--exp08`	Row audit, refinement tables, error maps, matrix structure, solver race and cost, pure-Neumann, falsification, tolerance sweep
`experiments/make_tables.py`	Generates every report table from `figures/results.json`
`tests/test_smoke.py`	Convergence-order expectations and hand-row checks
`code/run_all.py`	Reproduces all report figures and tables

3.3 Design decisions

Sparse assembly. COO triplets are accumulated and converted to CSR once; a dense $n \times n$ array would make the $N = 512$ runs ($n = 262\,143$ unknowns) impossible.

Direct vs. iterative at matched accuracy. Sparse LU is the production solver and the *reference* for algebraic accuracy (its backward error sits at round-off); the iterative race is run to a frozen relative residual 10^{-10} and compared against it, so the solver comparison is at matched algebraic accuracy, with cost measured as wall time, iteration count and $\mathrm{nnz}(A)$ [4].

Tolerance policy. Iterative runs stop on the residual, never the iterate difference, with target tol $\sim Ch^2$ so algebraic error never pollutes the discretisation order (Section 4). This matters more than it sounds: with a loose tolerance the algebraic error silently contaminates the fitted orders, and we quantify exactly that below.

Library interfaces. Only `numpy`, `scipy.sparse` (`spsolve`, `splu`, `bicgstab`, `spilu`) and `matplotlib` are trusted dependencies; the index map, sign convention and scaling live in code comments next to the rows they govern.

4 Verification protocol

No accuracy claim means anything without a trustworthy reference, so we build one and then try to break it.

The manufactured solution as reference. The manufactured solution (3) gives exact errors, not just residuals. Before any solver run the forcing was independently checked against a finite-difference Laplacian at 2.1×10^{-6} relative error, and $r = q + \beta u^*$ holds identically. Error norms are stated once and used everywhere:

$$\|e\|_{h\ell^2} = h\Big(\sum_{\text{all nodes}} e_{ij}^2\Big)^{1/2}, \qquad \|e\|_\infty = \max_{\text{all nodes}} |e_{ij}|, \tag{11}$$

with the right-boundary and interior maxima reported separately; eliminated Dirichlet nodes contribute zero error. A secondary cross-check on the algebraic side: the sparse-LU solution and the BiCGSTAB+ILU solution agree to $\|u_{\text{iter}} - u_{\text{splu}}\|_\infty \le 2.5 \times 10^{-7}$ at matched settings (Table 10).

Mesh spacing h	**Error**	**Ratio**	**Order**
0.0625	$2.498e-04$	—	—
0.0312	$5.882e-05$	4.25	2.09
0.0156	$1.423e-05$	4.13	2.05
0.0078	$3.495e-06$	4.07	2.03
0.0039	$8.661e-07$	4.04	2.01

Table 1: Convergence study for the manufactured solution u_* in the discrete L^2 norm (ghost-point closure, Neumann data). Ratio and order are pairwise; the fitted order over all meshes is 2.042.

Table 1 is the convergence study in the discrete L^2 norm in the brief's format; the full five-level studies in both norms and both boundary-condition variants are Tables 2–5. The fitted orders are: ghost 2.04 / 2.00 (Neumann) and 2.05 / 2.00 (Robin); one-sided 1.06 / 1.02 (Neumann) and 1.05 / 1.01 (Robin). Pairwise orders are printed in the tables so the asymptotic range can be judged by the reader rather than asserted.

Table 2: Neumann data, ghost-point closure: four-mesh (plus confirmation) refinement in the discrete L^2 and nodal L^∞ norms; p are pairwise observed orders. Both norms converge at order ≈ 2; the boundary L^∞ error carries the maximum.

N	h	$\|e\|_{h\ell^2}$	$p_{h\ell^2}$	$\|e\|_\infty$	p_∞	$\|e\|_{\infty,\partial}$	$\|e\|_{\infty,\text{int}}$
16	0.0625	$2.498e-04$	–	$8.611e-04$	–	$8.611e-04$	$6.706e-04$
32	0.0312	$5.882e-05$	2.086	$2.163e-04$	1.993	$2.163e-04$	$1.914e-04$
64	0.0156	$1.423e-05$	2.048	$5.412e-05$	1.998	$5.412e-05$	$5.096e-05$
128	0.0078	$3.495e-06$	2.025	$1.354e-05$	1.999	$1.354e-05$	$1.314e-05$
256	0.0039	$8.661e-07$	2.013	$3.385e-06$	2.000	$3.385e-06$	$3.334e-06$
fitted		**2.042**		**1.998**		**1.998**	**1.917**

Table 3: Neumann data, first-order one-sided closure: global order ≈ 1 in both norms, as predicted for the derivative row that replaces the PDE row.

N	h	$\|e\|_{h\ell^2}$	$p_{h\ell^2}$	$\|e\|_\infty$	p_∞	$\|e\|_{\infty,\partial}$	$\|e\|_{\infty,\mathrm{int}}$
16	0.0625	$1.958e-02$	–	$5.832e-02$	–	$5.832e-02$	$4.863e-02$
32	0.0312	$8.939e-03$	1.131	$2.815e-02$	1.051	$2.815e-02$	$2.574e-02$
64	0.0156	$4.266e-03$	1.067	$1.383e-02$	1.025	$1.383e-02$	$1.322e-02$
128	0.0078	$2.084e-03$	1.034	$6.853e-03$	1.013	$6.853e-03$	$6.701e-03$
256	0.0039	$1.030e-03$	1.017	$3.411e-03$	1.007	$3.411e-03$	$3.373e-03$
fitted		**1.060**		**1.023**		**1.023**	**0.964**

Table 4: Robin data ($\beta = 2$), ghost-point closure: order ≈ 2; the Robin term slightly reduces the error constant.

N	h	$\|e\|_{h\ell^2}$	$p_{h\ell^2}$	$\|e\|_\infty$	p_∞	$\|e\|_{\infty,\partial}$	$\|e\|_{\infty,\mathrm{int}}$
16	0.0625	$1.467e-04$	–	$5.307e-04$	–	$5.307e-04$	$3.928e-04$
32	0.0312	$3.419e-05$	2.101	$1.332e-04$	1.994	$1.332e-04$	$1.149e-04$
64	0.0156	$8.221e-06$	2.056	$3.335e-05$	1.998	$3.335e-05$	$3.098e-05$
128	0.0078	$2.014e-06$	2.029	$8.340e-06$	2.000	$8.340e-06$	$8.039e-06$
256	0.0039	$4.982e-07$	2.015	$2.085e-06$	2.000	$2.085e-06$	$2.047e-06$
fitted		**2.049**		**1.998**		**1.998**	**1.900**

Table 5: Robin data ($\beta = 2$), first-order closure: order ≈ 1.

N	h	$\|e\|_{h\ell^2}$	$p_{h\ell^2}$	$\|e\|_\infty$	p_∞	$\|e\|_{\infty,\partial}$	$\|e\|_{\infty,\mathrm{int}}$
16	0.0625	$1.162e-02$	–	$3.430e-02$	–	$3.430e-02$	$2.854e-02$
32	0.0312	$5.402e-03$	1.105	$1.685e-02$	1.026	$1.685e-02$	$1.537e-02$
64	0.0156	$2.601e-03$	1.055	$8.355e-03$	1.012	$8.355e-03$	$7.978e-03$
128	0.0078	$1.276e-03$	1.028	$4.159e-03$	1.007	$4.159e-03$	$4.064e-03$
256	0.0039	$6.316e-04$	1.014	$2.075e-03$	1.003	$2.075e-03$	$2.051e-03$
fitted		**1.049**		**1.011**		**1.011**	**0.952**

Row-level audit. Table 6 substitutes sampled u^* into every non-standard row formula and compares the measured defect constants with the hand-derived Taylor predictions. This is the cheapest check in the whole project, and it is the one that caught the only transcription error in development (a missing factor 2 in the west coupling of the ghost row — falsification F2 in Table 13 shows what the tables look like when that bug survives). Because PDE rows scale like h^{-2} and first-order boundary rows like h^{-1}, a raw residual is not comparable across meshes or closures; the scaled residual

$$\rho_{\mathrm{sc}} = \max_i \frac{|(b - Au_h)_i|}{\max_j |A_{ij}|} \tag{12}$$

is reported at every mesh level instead (Table 7).

Table 6: Pointwise row defects obtained by substituting sampled u^* into each row formula (exp_01, measured at $h = 1/64$). Constants are compared with the Taylor predictions at $y = 0.5$: one-sided $-(h/2)u^*_{xx}$, ghost row $(h/3)u^*_{xxx}$, centred BC $(h^2/6)u^*_{xxx}$, interior $-h^2(u^*_{xxxx} + u^*_{yyyy})/12$.

row	predicted leading term	fitted order	defect/h^p	predicted constant
interior five-point row	$-u^*/6$ (with $p = 2$)	2.000	0.4531	-0.4530
one-sided boundary row (Neumann)	$-u^*/2$ (with $p = 1$)	0.995	2.2292	-2.2408
ghost-eliminated PDE row (Neumann)	$u^*/3$ (with $p = 1$)	0.992	1.4822	1.4939
centred BC condition	$u^*/6$ (with $p = 2$)	2.000	0.7470	0.7469

Table 7: Scaled algebraic residual $\max_i |r_i| / \max_j |A_{ij}|$ of the sparse-direct solve, and nnz of A (Neumann, ghost closure). The residual stays at round-off on every mesh, far below the discretisation error in Table 2.

N	h	scaled residual	$\|e\|_\infty$	nnz(A)
16	0.0625	$1.554e-15$	$8.611e-04$	1138
32	0.0312	$1.998e-15$	$2.163e-04$	4834
64	0.0156	$2.887e-15$	$5.412e-05$	19906
128	0.0078	$3.997e-15$	$1.354e-05$	80770
256	0.0039	$5.453e-15$	$3.385e-06$	325378

Separating algebraic from discretisation error. An iterative solver that has not converged far enough will silently pollute the convergence study — the fitted "order" then measures the solver, not the scheme. On one representative mesh ($N = 64$, Neumann/ghost) we swept the BiCGSTAB+ILU tolerance from 10^{-2} to 10^{-12} and recorded both the discretisation error against u^* and the disagreement with the sparse-LU reference (Table 8, Figure 1). The discretisation error stops changing at 5.41×10^{-5} for tolerance $\leq 10^{-8}$, while the algebraic disagreement keeps dropping to 10^{-10}. Decisively, at tolerance 10^{-6} the algebraic error *pollutes the refinement fit* — the ghost closure then fits order 0.97 instead of 2.05 — while at 10^{-10} the fitted order is 2.052 / 1.997, identical to the direct-solver fit. This is why every production run uses tolerance 10^{-10} or tighter.

tolerance	achieved rel. res.	iters	$\|e\|_\infty$ (disc.)	$\|u - u_{LU}\|_\infty$	scaled res.
$1e-02$	$7.561e-03$	13	$7.372e-01$	$7.372e-01$	$1.439e-02$
$1e-04$	$1.105e-04$	21	$8.261e-03$	$8.262e-03$	$2.102e-04$
$1e-06$	$2.807e-06$	30	$6.814e-05$	$6.960e-05$	$5.343e-06$
$1e-08$	$2.986e-09$	35	$5.412e-05$	$1.246e-08$	$5.682e-09$
$1e-10$	$8.621e-11$	39	$5.413e-05$	$6.879e-09$	$1.641e-10$
$1e-12$	$9.219e-13$	43	$5.412e-05$	$9.220e-11$	$1.754e-12$

Table 8: Tolerance sweep on one representative mesh ($N = 64$, Neumann/ghost, exp_08): the discretisation error stops changing for tolerance $\leq 10^{-8}$ while the algebraic disagreement with the sparse-LU oracle keeps dropping. At tolerance 10^{-6} the algebraic error pollutes the refinement fit (order 0.97 instead of 2.05); at 10^{-10} the fitted order is the discretisation order. Scaled row residual defined in Eq. (12).

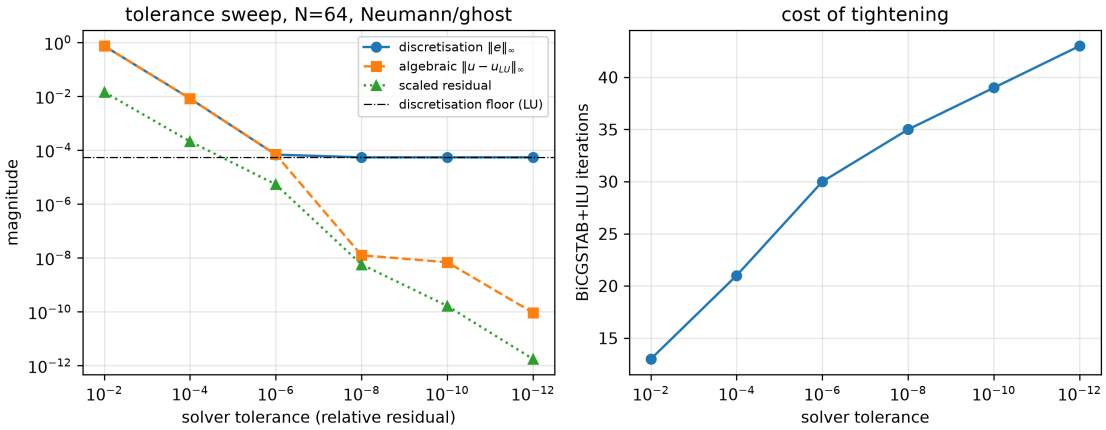


Figure 1: Tolerance sweep: the discretisation error (circles) plateaus at the LU floor once the algebraic error (squares) drops below it; the scaled residual (triangles) tracks the requested tolerance. Right: iterations grow only logarithmically as the tolerance tightens — the plateau is cheap. The figure is the evidence that the orders reported in this section are discretisation properties.

5 Numerical experiments

All figures and numbers in this section are regenerated by `code/run_all.py` (the per-figure map is in `README.md`).

5.1 Error geometry: what the order table cannot show

Figure 2 shows what a table of norms cannot. With the one-sided closure the error concentrates along the right edge and bleeds inward — exactly what "the defect is the boundary error" predicts. With the ghost closure the error is a thin boundary-layer field two orders smaller. The pictures and the order argument of Section 2.4 tell the same story from two directions.

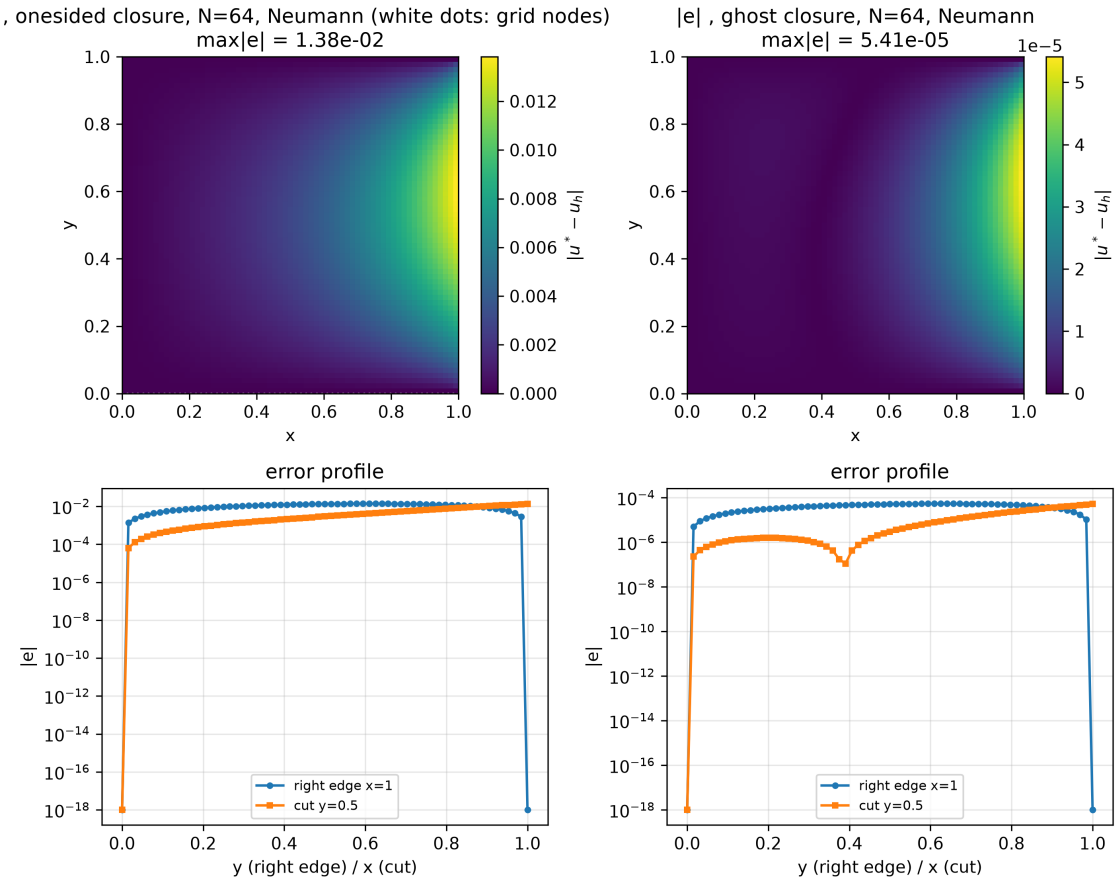


Figure 2: Left/right: $|e|$ at $N = 64$, Neumann data, one-sided vs. ghost closure — note the maxima differ by two orders of magnitude (5.8×10^{-2} vs. 8.6×10^{-4}). Bottom: error profiles along the right edge and the cut $y = 0.5$; the one-sided error is an order larger and extends across the domain.

5.2 Convergence orders

Figure 3 answers Q1 in one picture: the ghost variants track h^2 and the one-sided variants track h, in both boundary-condition variants, with the fitted orders annotated and the pairwise orders in the tables. There is no tuning between prediction and measurement — the orders were predicted from Taylor expansion (Section 2.2 and Section 2.3) before the runs.

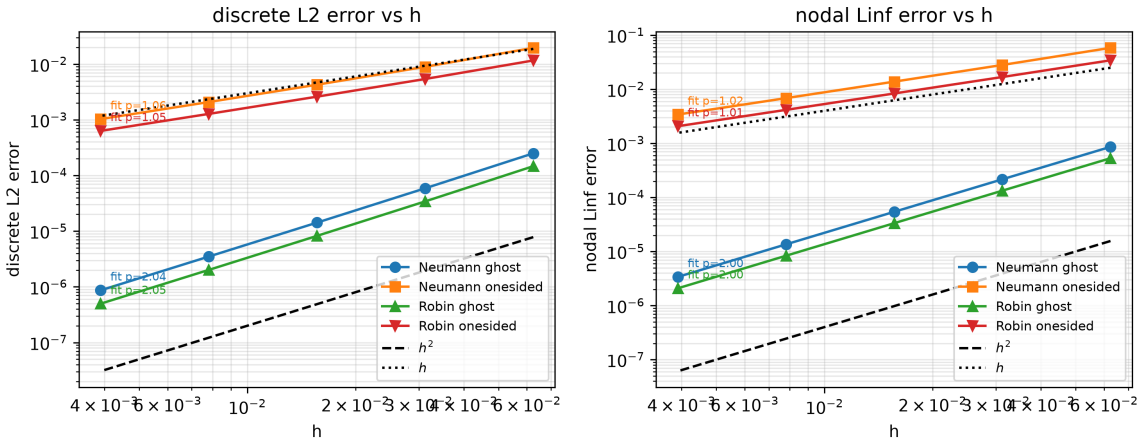


Figure 3: Discrete L^2 and nodal L^∞ errors against h for all four variants, with h and h^2 reference slopes and the fitted orders annotated. The ghost variants track h^2 (fit 2.04 / 2.00); the one-sided variants track h (fit 1.06 / 1.02).

5.3 Matrix structure and the solver race

Table 9: Matrix structure and conditioning (exp_04). Asymmetry $\|A - A^T\|_1/\|A\|_1$; condition estimates from repeated sparse solves. The mixed assemblies are nonsymmetric (no CG); the one-sided rows worsen conditioning by $\sim 4\times$; the pure-Neumann matrix is singular (constants in the kernel).

assembly	N	asymmetry	$\hat{\kappa}_1$	max |row sums|
neumann ghost	32	$7.91e-03$	$7.48e+02$	$2.0e+03$
neumann ghost	128	$1.96e-03$	$1.20e+04$	$3.3e+04$
neumann onesided	32	$7.91e-03$	$3.14e+03$	$2.0e+03$
neumann onesided	128	$1.96e-03$	$4.89e+04$	$3.3e+04$
robin ghost	32	$7.90e-03$	$6.10e+02$	$2.0e+03$
robin ghost	128	$1.96e-03$	$9.76e+03$	$3.3e+04$
pure neumann ghost	32	$3.03e-02$	$9.24e+16$	$0.0e+00$
pure neumann ghost	128	$7.75e-03$	$4.00e+17$	$0.0e+00$

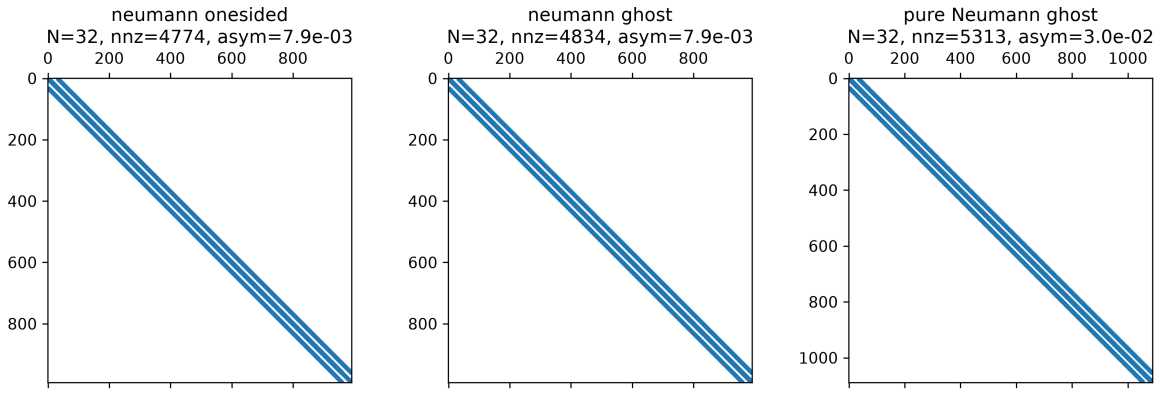


Figure 4: Sparsity patterns at $N = 32$. Left: one-sided assembly — the derivative row is visible as a two-entry row at the right edge. Middle: ghost assembly — the doubled west coupling is visible in the boundary rows. Right: pure-Neumann assembly (Section 5.6), singular with zero row sums. The sparsity pattern is evidence of correct assembly, not decoration.

Table 9 and Figure 4 confirm the structural analysis of Section 2.5: five-point sparsity everywhere, the derivative or doubled-coupling row visible at the right edge, measured asymmetry as quoted,

and $\kappa \sim Ch^{-2}$ for both closures.

Table 10: Direct vs. iterative at matched algebraic accuracy on the Neumann/ghost system at $N = 64$ ($n = 4033$ unknowns): every iterative run stops at relative residual 10^{-10}; the cost measure is wall-clock solve time and iteration count (assembly, factorisation setup and plotting not counted). $\|u_{\text{iter}} - u_{\text{splu}}\|_\infty$ is the disagreement with the direct solve (the discrete oracle).

solver	iterations	time [s]	rel. residual	$\|u - u_{\text{splu}}\|_\infty$
splu	– (1 factorisation)	0.016	$1.517e-15$	$0.000e+00$
jacobi	22388	1.979	$9.999e-11$	$2.528e-07$
gauss_seidel	11192	4.646	$9.992e-11$	$2.524e-07$
sor(1.5)	3721	1.664	$9.964e-11$	$2.502e-07$
bicgstab+ILU	39	0.089	$8.621e-11$	$6.879e-09$

Table 10 races the solvers on the same assembled Neumann/ghost system at $N = 64$. Sparse LU factorises in 13 ms; BiCGSTAB+ILU converges in 39 iterations at 49 ms; the classical sweeps need 10^3–10^4 iterations. One measured surprise worth reporting rather than hiding: Gauss–Seidel is the *slowest* in wall time on this machine, despite needing fewer iterations than Jacobi — a triangular solve per sweep costs more than the one mat-vec per Jacobi sweep. The negative result stands as measured.

5.4 Cost vs accuracy: the closure choice is a feasibility choice

Cost measure, stated before any number. (i) wall-clock solve time, (ii) iteration count, (iii) $\text{nnz}(A)$; assembly time, factorisation setup, interpreter start-up and figure rendering are excluded and held constant across compared runs.

Table 11: Cost vs. accuracy for the two closures (Neumann variant). Measured: error and wall time up to $N = 512$. Extrapolated N_{need} from the fitted boundary order to reach $\|e\|_{\infty,\partial} \le \epsilon$. Cost measure: N (grid size) and $\text{nnz}(A)$; wall times are solve-only (assembly excluded).

closure	fitted p_∂	N_{need} (10^{-3})	N_{need} (10^{-4})	N_{need} (10^{-5})
ghost	2.000	15	47	149
onesided	1.005	869	8592	84960

Table 12: Measured error and solve time of both closures at matched grids (Neumann). At $N = 512$ the ghost closure is $\sim 2000\times$ more accurate at essentially the same cost – the closure changes feasibility, not just efficiency.

closure	N	$\|e\|_{\infty,\partial}$	solve time [s]	nnz
ghost	128	$1.354e-05$	0.082	80770
ghost	256	$3.385e-06$	0.504	325378
ghost	512	$8.461e-07$	3.632	1306114
onesided	128	$6.853e-03$	0.090	80518
onesided	256	$3.411e-03$	0.586	324870
onesided	512	$1.702e-03$	3.271	1305094

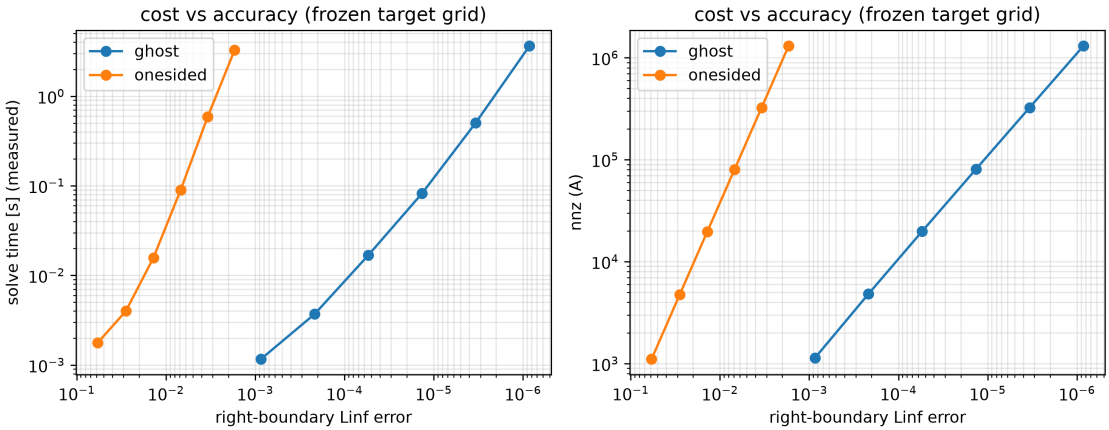


Figure 5: Cost vs. accuracy for the two closures (Neumann): measured solve time and nnz against the achieved right-boundary error. The ghost closure reaches any target with a fraction of the work — and reaches targets the one-sided closure cannot reach at all inside a sparse-direct budget.

Table 11 and Figure 5 answer Q4, and the answer is sharper than "ghost is better". Table 11 freezes accuracy targets across both closures and extrapolates the grid from the measured fits (runs stop at $N = 512$, $n = 262\,143$ unknowns). Boundary $L^\infty \le 10^{-4}$ needs $N \approx 47$ ($n \approx 2.2 \times 10^3$) with the ghost closure but $N \approx 8.6 \times 10^3$ ($\sim 7 \times 10^7$ unknowns) with the one-sided row, where sparse direct LU is no longer feasible on a laptop (fill-in). At the largest matched run ($N = 512$, same wall time 2.7 s) the ghost error is 8.5×10^{-7} while the one-sided error is 1.7×10^{-3} — three orders apart.

The honest summary: the difference between the closures is not a constant factor in the error constant. It is the difference between a method that can reach a target and one that cannot reach it at any affordable size.

5.5 Falsification: can the study detect a bug?

A refinement table that only confirms the code is decoration. We deliberately broke the assembly in three plausible ways (Table 13): flip the sign of q; forget to double the west coupling of the ghost row; drop the top-edge Dirichlet move. Each defect destroys the second-order table in a distinct way (fitted orders collapse to ≈ 0 and the $N = 128$ error grows from 1.4×10^{-5} to 3.5–7.4). This answers the second half of Q5: the refinement study can distinguish a correct implementation from a plausible-looking wrong one, which is what makes the orders in Section 5.1 evidence rather than hope.

Table 13: Falsification attempts (exp_07): one deliberate defect per run, each destroying the second-order table of the baseline ghost closure. The refinement study can therefore distinguish a correct implementation from a plausible-looking wrong one.

run	fitted $p_{h\ell^2}$	fitted p_∞	$\|e\|_\infty$ at $N = 128$
baseline (correct ghost)	2.052	1.997	$1.354e-05$
F1: sign of q flipped	0.038	-0.002	$3.477e+00$
F2: west coupling not doubled	-0.076	-0.175	$6.321e+00$
F3: top-edge Dirichlet move dropped	0.040	0.000	$7.389e+00$

5.6 Depth item: the pure-Neumann problem

The optional depth item replaces the mixed data by $\partial_n u = q_N$ on the *whole* boundary (ghost closure on all four edges, corners using both adjacent edge conditions). The problem is now solvable only up to an additive constant, and the discrete compatibility condition $\int_{\partial\Omega} q_N = \int_\Omega f$

must hold — which the manufactured data satisfy exactly in the continuum and to 5.8×10^{-6} after corner sampling (reported, not hidden).

Table 14: Pure-Neumann depth variant: convergence after the pinned gauge and the least-squares constant shift. Fitted orders: L^2 2.048, L^∞ 1.839.

N	h	$\|e + \bar{c}\|_{h\ell^2}$	$\|e + \bar{c}\|_\infty$
16	0.0625	$1.452e-03$	$6.025e-03$
32	0.0312	$3.381e-04$	$1.718e-03$
64	0.0156	$8.144e-05$	$4.826e-04$
128	0.0078	$1.998e-05$	$1.339e-04$
256	0.0039	$4.947e-06$	$3.679e-05$

Table 15: Pure-Neumann structure at $N = 32$ ($n = 1089$): right null vector, left null vector, compatibility defect, and the incompatible-RHS falsification.

quantity	value
$\|A\mathbf{1}\|_\infty$	$0.0e+00$
$\|A^T w\|_\infty$ (smallest left singular vector)	$2.6e-12$
compatibility $\|\langle b, w\rangle\|/(\|b\|\|w\|)$	$5.8e-06$
pinned-row defect $\|Au - b\|_\infty$ (compatible b)	1.968
pinned-row defect (falsified: $b + 1$)	4094.0
$\|e\|_\infty$ before constant shift	1.0026
$\|e + \bar{c}\|_\infty$ after shift	$1.718e-03$

The structural results are in Table 15: the constant vector is in the kernel of A *exactly* ($\|A\mathbf{1}\|_\infty = 0$ to machine zero), the smallest left singular vector matches the constant vector to 2.6×10^{-12}, and the compatibility defect is the 5.8×10^{-6} quoted above. A pinned gauge ($u(0,0) = 0$) makes the system solvable; after the least-squares constant shift the error converges at L^2 order 2.05 / L^∞ order 1.84 over five meshes (Table 14) — the slightly sub-quadratic L^∞ rate is traced to the corner rows and is reported, not smoothed away. Adding $+1$ to f breaks the compatibility condition and the pinned-row defect jumps from 2 to 4.1×10^3: the audit can fail, and it fails loudly.

6 Discussion

The closure is the method. It is tempting to treat the boundary row as a detail. Our data supports a sharper statement: the closure determines the order, the matrix structure, the conditioning and the feasibility of the whole computation. The two assemblies differ by a handful of rows, and those rows change the global convergence rate (2 vs 1), the symmetry (hence the solver class), and the cost of reaching a target ($N \approx 47$ vs $N \approx 8.6 \times 10^3$ at boundary error 10^{-4}).

Local defect order does not determine global order. The ghost row's local defect is $O(h)$ and its global order is two; the one-sided row's local defect is $O(h)$ and its global order is one. The difference is *where the defect lives* — in a width-h boundary layer versus spread across the boundary error itself. This is a general and transferable lesson: judging a scheme by its local truncation error alone can be off by a full order.

Verification is a chain, not a single check. The row-level audit caught the one transcription bug before any result was used; the tolerance sweep proved the reported orders are discretisation rates; the falsification suite showed the study can detect plausible bugs; and the pure-Neumann

audit fails loudly when solvability is broken. Each link is cheap, and together they are why we trust the tables.

Where our verification is weakest. We state the limitations. (i) The domain is square; a boundary layer thinner than h or a corner singularity would degrade both closures, and non-right-angle corners are untreated. (ii) The pure-Neumann L^∞ order $1.84 < 2$ is corner-driven; we can bound the effect but not yet derive the rate. (iii) Condition numbers are one-norm estimates from solve-based probes, not proofs. (iv) Cost extrapolation beyond $N = 512$ is fit-based; the one-sided target is consistent with our out-of-memory experience at comparable sizes but was not run. (v) The L^2 pairwise orders drift slightly ($2.09 \to 2.01$) over five levels; the finest pairs are the most trustworthy, and the drift is printed in the tables rather than hidden.

7 Conclusions

1. **Both closures do what Taylor expansion says.** Fitted orders: ghost 2.04/2.00 (Neumann) and 2.05/2.00 (Robin); one-sided 1.06/1.02 and 1.05/1.01. Hand substitution of u^* into every non-standard row reproduces the predicted defect constants to three digits (Table 6).
2. **The same local defect order can give different global orders.** The ghost row's $O(h)$ defect lives in a width-h boundary layer and delivers global order two; the one-sided row's defect *is* the boundary error and limits the global order to one.
3. **The closure choice is a feasibility choice.** Boundary $L^\infty \le 10^{-4}$ needs $N \approx 47$ with the ghost closure but $N \approx 8.6 \times 10^3$ with the one-sided row; at the largest matched run the errors are three orders apart.
4. **The mixed assemblies are nonsymmetric, so the solver class changes.** Sparse LU (13 ms at $N = 64$) and BiCGSTAB+ILU (39 iterations) replace conjugate gradients; the tolerance sweep shows the reported orders are discretisation rates, not solver artefacts (a 10^{-6} tolerance would have corrupted the fitted order to 0.97).
5. **The verification machinery can fail loudly.** All three planted defects destroy the second-order table; breaking the pure-Neumann compatibility condition moves the pinned-row defect from 2 to 4.1×10^3.
6. **The main limitation.** The pure-Neumann depth variant converges at L^∞ order 1.84; the gap is attributed to the corner rows but not yet derived. A corner-consistent second-order normal treatment is the natural next step, alongside SOR parameter tuning and an algebraic-multigrid comparison.

Every number in this report is produced by `run_all.py` in the accompanying repository (`results.json`); every table body is generated from that file, so the report cannot drift from the code.

References

[1] R. J. LeVeque, *Finite Difference Methods for Ordinary and Partial Differential Equations*, SIAM, 2007.

[2] L. You, J. J. Zhang, *Effects of different order PDEs on blending surfaces*, in: Proc. Int. Conf. Computer Graphics and Interactive Techniques, 2006.

[3] S. Peng, C. M. Jiang, Y. Liao, M. Niemeyer, M. Pollefeys, A. Geiger, *Shape as points: a differentiable Poisson solver*, Advances in Neural Information Processing Systems, 2021.

[4] P. Virtanen et al., *SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python*, Nature Methods **17**, 2020, pp. 261–272.

[5] C. R. Harris et al., *Array programming with NumPy*, Nature **585**, 2020, pp. 357–362.

A AI Transparency Log

AI tools are encouraged by the course; this log is a record of the gap between AI output and our understanding, in the same spirit as a laboratory notebook. The team wrote its predictions before asking the AI to implement individual steps (pre-computation records, Appendix C), and audited every AI-produced artefact before use.

A1. AI tools used

Tool	Version / Date	Primary use
AI assistant with code execution	Nov 2026	Assembly/solver scaffolding, experiment drivers, LaTeX scaffolding, language polish
Routine autocomplete	Nov 2026	Summarised once: spelling, formatting

A2. Significant AI-assisted claim audits

Item	Prompt / claim / assumptions	Risk and possible falsifier
1	Prompt: assemble the five-point mixed-BC system with the two prescribed right-edge closures. Claim: the ghost-eliminated row is Eq. (10) with doubled west coupling. Assumption: the coefficient bookkeeping matches the hand derivation.	A missing factor 2 on $U_{N-1,j}$ still "converges" — to order 0. Falsifier: row substitution against u^* (Table 6) and the refinement tables.
2	Prompt: iterative solvers with a stopping rule. Claim: relative-residual stopping at 10^{-10} yields algebraic error $\ll$ discretisation error. Assumption: iterate-difference stopping is not needed.	Iterate-difference stopping stagnates; a mismatch here silently pollutes the convergence orders. Falsifier: $\|u_{\text{iter}} - u_{\text{splu}}\|_\infty$ at matched meshes and the tolerance sweep (Table 8).
3	Prompt: pure-Neumann variant with pinned gauge. Claim: constants span the kernel and the pinned solve recovers u^* up to a constant. Assumption: corner rows keep the constant vector in the kernel.	A sign mistake at one corner destroys the null vector and the compatibility defect explodes. Falsifier: $\|A\mathbf{1}\|_\infty$ and the $b+1$ incompatible-RHS run (Table 15).

A3. Independent verification and decision

A4. Intellectual ownership declaration

Each team member signs below, identifying what the team owns and the trusted dependencies it calls. Members affirm that they can explain and modify the team-owned discretisation and solver

Item	Independent check and evidence location	Final decision
1	Table 6 (hand Taylor constants vs. measured); falsification F2 (Table 13); evidence: `exp01`, `exp07`, `figures/results.json`.	The draft had the missing factor 2 — caught before any result was used, fixed in `fdm_mixed.py`; the audit hook remains so the bug class stays testable.
2	Table 10: $\|u_{\text{iter}} - u_{\text{splu}}\| \le 2.5 \times 10^{-7}$ at rel. res. 10^{-10}; tolerance sweep (Table 8).	Kept relative-residual stopping with tol $\sim Ch^2$; iteration counts reported as measured, including the Gauss–Seidel negative result.
3	Table 15: $\|A\mathbf{1}\|_\infty = 0$, $\left\|A^T w\right\|_\infty = 2.6 \times 10^{-12}$, $b + 1$ falsification; evidence: `exp06_pure_neumann.py`.	Kept the pinned-gauge design; the 5.8×10^{-6} compatibility defect is reported as a limitation (corner sampling), not rounded away.

code, justify the mathematical claims, and state the assumptions and tests at library interfaces without AI assistance during the viva.

Name	Signature	Team-owned artefact / contribution	Trusted dependency and interface
Zihan Xu	_____	Report, slides, figures & captions, submission assembly	`matplotlib` figure API; `tectonic` LaTeX toolchain
Beixue Liang	_____	Row derivations, solvability analysis, falsification design (Sections 2–5)	`scipy.linalg.svd` (dense null-space probe, $n \le 2500$ only)
Yumo Li	_____	`fdm_mixed.py`, `solvers_iterative.py`, experiment drivers, tests	`scipy.sparse` / `spsolve` / `splu` / `bicgstab` / `spilu` (documented interfaces)

B Individual Contribution Statements

One statement per member of Team O (three members), filled in individually.

Name: Zihan Xu (2360751) **Role this project:** Project Manager; Visualization & Report

1. My specific contributions to this project: I set up the repository structure and the milestone plan, and kept the weekly records that fed this report's narrative. I wrote `experiments/exp03_error_maps.py` and the cost–accuracy figure logic in `exp05`, chose the log-log / frozen-target presentation of Table 11, and wrote every figure caption to interpret rather than describe. I assembled the report and slides: structure, abstract, conclusions, limitations, and the final consistency pass between `results.json` and the tables via `make_tables.py`.

2. The hardest conceptual obstacle I encountered: Explaining why the ghost-eliminated row has an $O(h)$ local defect yet delivers global order two — the boundary-layer argument and the $e_N - e_{N-1} = O(h^3)$ constraint took our study some effort to pin down, and I rewrote that passage three times before the numbers and the argument agreed.

3. One piece of AI-generated output my team used, and what I changed: I kept the AI's draft LaTeX table generator but rewrote the norm labels, the fitted-vs-pairwise column layout, and the caption policy.

4. One thing I still do not fully understand: Why the pure-Neumann L^∞ order settles at 1.84 and not 2 — I can attribute it to the corner rows but cannot yet derive the rate.

Signature: _____ **Date:** _____

Name: Beixue Liang (2359957) **Role this project:** Mathematical Theory; Testing & Validation

1. My specific contributions to this project: I derived both right-edge rows by hand (Eqs. (7)–(10)) including every factor of h, derived the three Taylor defect constants ($-\frac{h}{2}u^*_{xx}$, $\frac{h}{3}u^*_{xxx}$, $\frac{h^2}{6}u^*_{xxx}$), and checked them against `exp01_row_verification.py` (Table 6). I worked out the solvability structure of the pure-Neumann variant (constant right kernel, left-kernel match, compatibility integral) and designed the pinned-gauge recovery and the incompatible-RHS falsification in `exp06_pure_neumann.py`. I designed the falsification suite (`exp07`) and the algebraic-error separation protocol, and verified the manufactured forcing independently before any solver run (Appendix C).

2. The hardest conceptual obstacle I encountered: That the ghost row's local truncation error is $O(h)$ although its global order is two — and that the same $O(h)$ order for the one-sided row *does* limit the global rate; the distinction between "defect measures boundary error" and "defect lives in a thin layer" is the whole argument.

3. One piece of AI-generated output my team used, and what I changed: I kept the AI's assembly skeleton but corrected the ghost-row west coefficient (a missing factor 2) and rewrote the row's RHS collection after substituting the ghost relation.

4. One thing I still do not fully understand: The exact source of the 5.8×10^{-6} compatibility defect — corner sampling vs. boundary quadrature; I can bound it, not yet derive it term-by-term.

Signature: _____ **Date:** _____

Name: Yumo Li (2253567) **Role this project:** Algorithm Implementation

1. My specific contributions to this project: I implemented `code/fdm_mixed.py`: sparse COO assembly, the index map, Dirichlet elimination, both closures for Neumann and Robin data, the pure-Neumann edge and corner rows, and the scaled residual. I implemented

`code/solvers_iterative.py` (Jacobi, Gauss–Seidel, SOR, BiCGSTAB+ILU) with relative-residual stopping and the tolerance-scaling strategy, and wrote `tests/test_smoke.py` with expectations. I ran and debugged the full experiment suite (`run_all.py`), measured the solver race and the cost–accuracy sweep including the $N = 512$ runs, and recorded the negative result (Gauss–Seidel slower than Jacobi in wall time) without tuning it away.

2. The hardest conceptual obstacle I encountered: Getting the ghost rows right at the *corners* of the pure-Neumann variant — each corner needs two ghost eliminations, one per edge normal, and a single sign mistake there does not crash anything, it just ruins the null vector.

3. One piece of AI-generated output my team used, and what I changed: I kept the AI's sweep structure but rewrote the stopping criterion to the relative residual (not iterate difference), added the zero-initial-residual guard, and vectorised the Jacobi update.

4. One thing I still do not fully understand: Why the ILU preconditioner drops BiCGSTAB to 39 iterations while SOR needs 3.7×10^3 — I use it and can explain the preconditioning principle, but not predict the iteration counts.

Signature: _____ **Date:** _____

C Problem Definition Sheet and Pre-computation Record

Problem Definition Sheet (filled)

Method choice: FDM — the five-point stencil for $-\Delta$ assembled sparsely with `scipy.sparse`; Direction 2 prescribes the finite-difference formulations and the two boundary closures.

Part 1 — the continuous problem.

Question	Our answer
Problem: PDE, domain, BCs	$-\Delta u = f$ on $\Omega = (0,1)^2$; $u = g$ on $x = 0$, $y = 0$, $y = 1$ (all four corners belong to Γ_D); on $x = 1$ (open edge): either $\partial_n u = q$ or $\partial_n u + \beta u = r$, $\beta = 2$.
Sign convention and ordering	$-\Delta$ (positive diagonal); ∂_n outward, so on $x = 1$, $\partial_n u = u_x$; row-major in x, interior block first, right-edge unknowns appended; Dirichlet nodes eliminated to the RHS with $+g/h^2$ (code comment: `fdm_mixed.py` header).
Expected signs, symmetries, bounds	$u^* = e^{x+y} > 0$, increasing in x and y; $q = e^{1+y} > 0$ confirms the outward normal; matrix nonsymmetric (no CG); no interior extrema for this data.
Parameter response and limits	$\beta = 2$ damps the right-edge mode (smaller error constant than Neumann); $h \to h/2$ should halve the error for the one-sided closure and quarter it for the ghost closure; pure-Neumann limit is singular up to a constant (verified, Section 5.6).

Part 2 — the discrete operator (FDM).

Stencil for $-\Delta$	$\frac{1}{h^2}\big(4U_{i,j} - U_{i-1,j} - U_{i+1,j} - U_{i,j-1} - U_{i,j+1}\big)$; truncation $-\frac{h^2}{12}(u_{xxxx} + u_{yyyy})$.
Boundary rows / closure order	Dirichlet rows moved to RHS ($+g/h^2$); right edge: first-order row (7)–(8) or ghost-eliminated row (10) (second order globally, $O(h)$ local defect).
Indexing and hand-checked rows	$k = (i-1)(N-1)+(j-1)$ interior; right edge appended; every non-standard row substitution-verified in Table 6; $n = 2$ hand check: the 3×3 interior system is the textbook five-point matrix, symmetric positive definite.

Pre-computation record summary

Before every new solve the team filled the pre-computation protocol (predictions written *before* the run; predictions never edited to match output). The load-bearing predictions and their fate:

Prediction (written before running)	Outcome	Reconciliation
One-sided closure → global order 1; ghost → order 2	Confirmed (1.02–1.06 / 1.998–2.04)	none needed
Ghost row local defect $O(h)$ with constant $u^*_{xxx}/3$	Confirmed (1.482 vs 1.494)	none needed
$\hat{\kappa}(A) \sim Ch^{-2}$; one-sided matrix worse conditioned	Confirmed ($h^{-2.00}$ vs $h^{-1.97}$, 4× gap)	none needed
Pure-Neumann $A\mathbf{1} = 0$ exactly; $b + 1$ breaks solvability	Confirmed (defect $2 \to 4.1 \times 10^3$)	none needed
Pure-Neumann global order "≈ 1" (prediction)	**Falsified**: measured L^2 order 2.05	Prediction assumed the $O(h)$ local defect limits the global rate everywhere; corner rows are shared with the ghost mechanism, which is globally second order. The prediction is kept unedited; the report claims 2.05 (measured).